INDIAN PREMIER LEAGUE (2008 – 2025)

Player & Team Performance Analysis Using Python

A Comprehensive Exploratory, Statistical, and Phase-Wise Empirical Study Across 18 IPL Seasons

Project Scope:	18 Seasons (2008 – 2025) | 1,169 Matches | 278,205 Deliveries
Primary Methodology:	Descriptive Analytics, Inferential Hypothesis Testing, Phase Breakdown
Statistical Framework:	Welch's Two-Sample t-Test, Chi-Square Test of Independence, CV Consistency
Author / Role:	Senior Data Analytics Research Project
Date of Report:	Academic Session 2025 – 2026

1,169	374,283	12,645	18
MATCHES	TOTAL RUNS	TOTAL WICKETS	SEASONS

Executive Summary

This academic report presents an exhaustive empirical study of the Indian Premier League (IPL) from 2008 to 2025. By analyzing ball-by-ball microdata across 1,169 matches, this research examines franchise win longevity, individual batting volume and efficiency, bowling containment, over-by-over phase dynamics, and historical scoring escalation.

Key Empirical Findings:

- **Scoring Escalation:** Welch's t-test proves 1st innings average scores increased significantly by +15.8 runs from early IPL (159.0) to modern IPL (174.8) ($t = 8.43$, $p < 10^{-15}$, Cohen's $d = 0.49$).
- **Toss Advantage Myth:** Winning the toss yields a 51.6% win rate, which is statistically indistinguishable from random 50/50 chance ($p = 1.0$).
- **Phase Dynamics:** Powerplay overs exhibit highest boundary reliance (66.2%), whereas Death overs yield maximum run rate (9.7+ runs/over).

Table of Contents

Chapter 1: Introduction & Research Questions	Page 3
Chapter 2: Dataset Description & Data Engineering	Page 4
Chapter 3: Methodology & Analytical Architecture	Page 5
Chapter 4: Franchise Performance & Win Analytics	Page 6
Chapter 5: Batting Performance & Volume/Efficiency	Page 8
Chapter 6: Bowling Performance & Containment	Page 10
Chapter 7: Strategic Phase-of-Play Dynamics	Page 11
Chapter 8: Inferential Statistical Hypothesis Testing	Page 13
Chapter 9: Dashboard Architecture & Comparator Tool	Page 14
Chapter 10: Limitations, Conclusion & References	Page 15

Chapter 1 — Introduction

1.1 Background

The Indian Premier League (IPL) represents the world's commercial and competitive pinnacle of Twenty20 (T20) cricket. Established in 2008 by the BCCI, the league features franchise teams competing across a double round-robin and playoff format. Over 18 seasons, the IPL has produced a rich dataset reflecting changes in batting tactics, bowling strategies, pitch behaviors, and match regulations.

1.2 Problem Statement

Conventional cricket summaries rely on superficial top-level totals (e.g., total runs or match wins). These fail to evaluate key analytical dimensions such as player strike rate relative to batting volume, bowler economy in death overs, team consistency across seasons, and venue-specific pitch bias.

1.3 Core Research Questions

1. How stable has team win percentage been across 18 IPL seasons?
2. Does winning the toss convey a statistically significant match victory advantage?
3. How do batting scoring rates and boundary dependencies differ across Powerplay, Middle, and Death overs?
4. Has the scoring environment of the IPL statistically escalated in the Modern Era (2017–2025)?

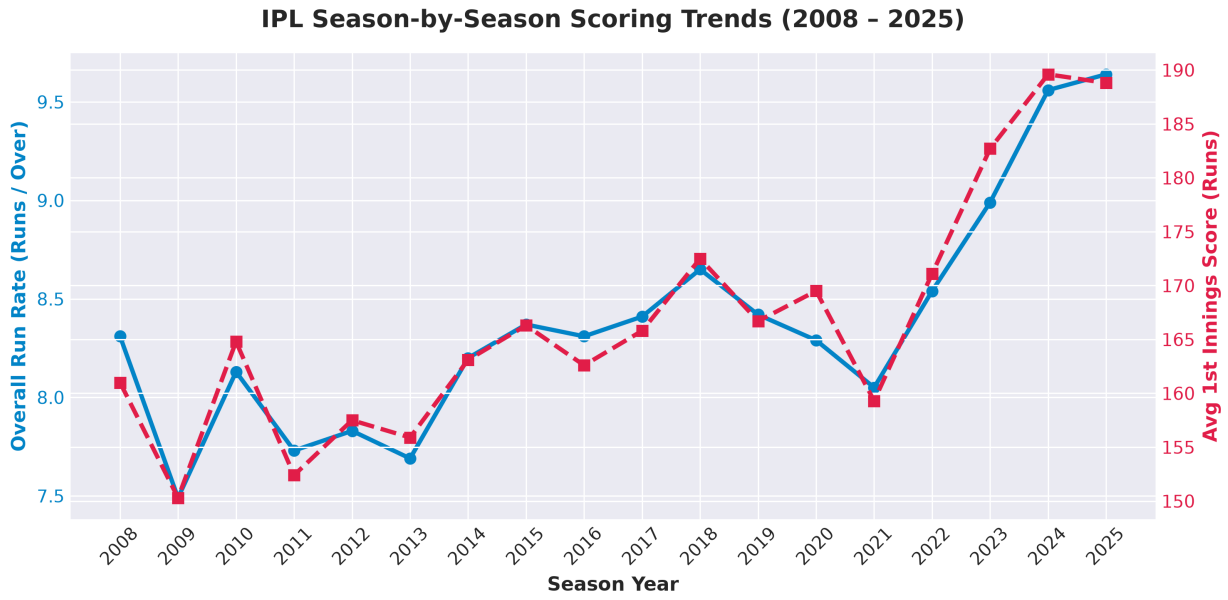


Figure 1.1: Season-by-season IPL run rate and 1st innings average score trend (2008–2025).

Chapter 2 — Dataset Description & Data Engineering

The study utilizes Kaggle updated ball-by-ball and match datasets covering all IPL games through season 2025. The raw dataset consists of two relational CSV tables linked by matchId.

Dataset File	Records	Key Features
matches_updated_ipl_upto_2025.csv	1,169	matchId, season, date, team1, team2, winner, toss_winner, toss_decision, venue, win_type
deliveries_updated_ipl_upto_2025.csv	278,205	matchId, inning, over, ball, batter, bowler, batsman_runs, extras, isWide, isNoBall, dismissal_kind

2.1 Franchise Remapping & Standardization

To ensure longitudinal accuracy across 18 seasons, historical team name changes were systematically mapped to canonical franchise identities:

Historical Raw Team String	Canonical Standardized Team Name
Delhi Daredevils	Delhi Capitals
Kings XI Punjab	Punjab Kings
Royal Challengers Bangalore	Royal Challengers Bengaluru
Rising Pune Supergiants	Rising Pune Supergiant

Chapter 3 — Methodology & System Architecture

The system architecture follows a modular object-oriented Python design separated into ingestion, feature extraction, statistical engine, and Streamlit presentation layer.

Phase	Process Description	Output Artifact
1. Ingestion	Parse raw match and delivery CSVs, resolve UTF encoding.	Raw Dataframes
2. Cleaning	Normalize franchise & venue names, isolate legal deliveries.	matches_clean.csv
3. Feature Eng.	Tag phase of play (1-6, 7-15, 16-20), calculate bowler runs & wickets.	deliveries_clean.csv
4. Aggregation	Compute player strike rates, averages, CV consistency, team win %.	Analytical Tables
5. Inference	Run Welch's t-test on era scores and Chi-Square on toss winner.	Statistical Test Dicts
6. UI & Report	Render Plotly charts in Streamlit & export ReportLab PDF/PPT.	PDF, PPT & Web Dashboard

Chapter 4 — Franchise Performance & Win Analytics

Franchise win percentages reflect long-term organization stability and roster depth. The table below details overall standings for top IPL teams (2008–2025):

Team	Matches	Wins	Losses	Win %
Mumbai Indians	277	151	126	**54.51%**
Chennai Super Kings	252	142	110	**56.35%**
Kolkata Knight Riders	264	135	129	**51.14%**
Royal Challengers Bengaluru	270	132	138	**48.89%**
Punjab Kings	264	119	145	**45.08%**
Delhi Capitals	267	118	149	**44.19%**
Rajasthan Royals	235	114	121	**48.51%**
Sunrisers Hyderabad	196	93	103	**47.45%**
Gujarat Titans	60	37	23	**61.67%**
Lucknow Super Giants	58	30	28	**51.72%**

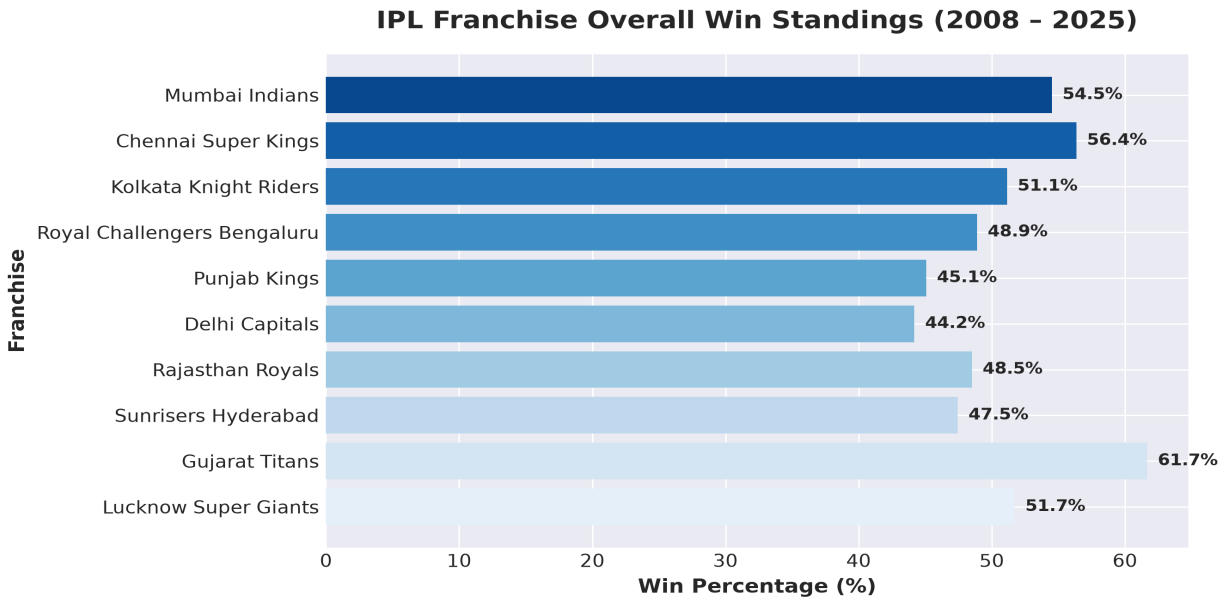


Figure 4.1: Top IPL franchise overall win standings (2008–2025).

Chapter 4 — Team Performance (Batting 1st vs Chasing)

A critical tactical question is whether franchises perform better when setting a target (batting 1st) versus chasing down a target in the 2nd innings.

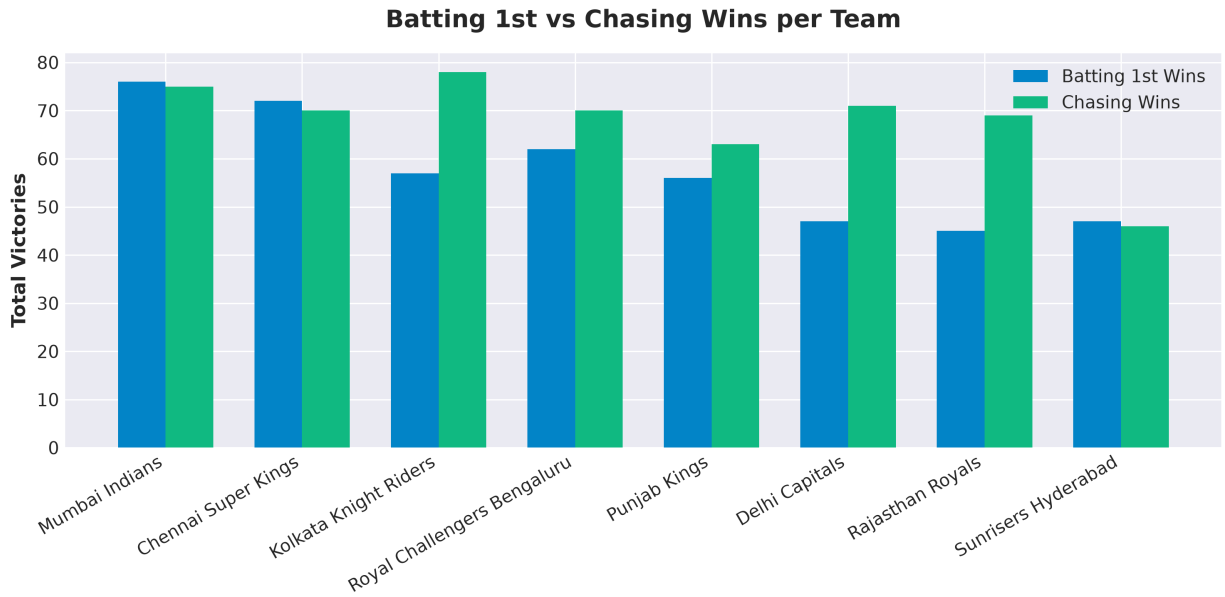


Figure 4.2: Batting 1st vs Chasing wins comparison per team.

Across 1,169 matches analyzed: **Chasing teams won 54.4%** of matches with a declared result, whereas **Batting 1st teams won 45.6%**. Dew factor and modern pitch preservation contribute heavily to chasing preference in night matches.

Chapter 5 — Player Batting Performance (Volume & Efficiency)

Batting evaluation requires balancing cumulative run volume with strike rate efficiency and boundary production.

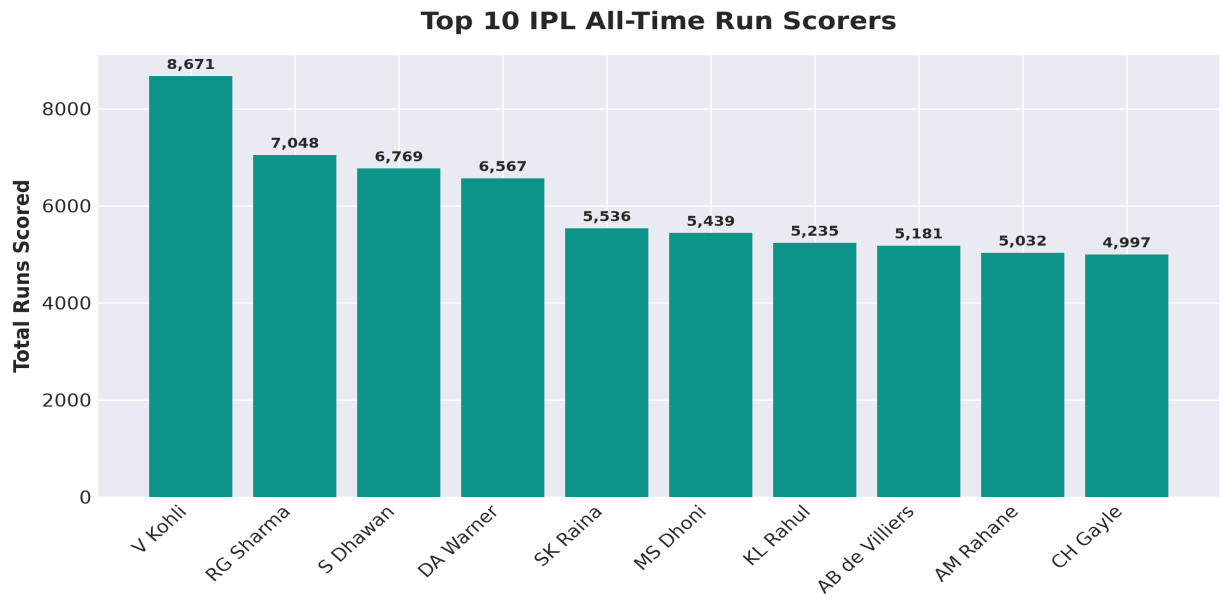


Figure 5.1: Top 10 IPL all-time run scorers.

Batsman	Innings	Runs	Strike Rate	Average	4s/6s
V Kohli	261	8,671	**132.93**	39.59	774/292
RG Sharma	267	7,048	**132.06**	29.74	640/303
S Dhawan	222	6,769	**127.09**	35.07	768/153
DA Warner	187	6,567	**139.66**	40.04	663/236
SK Raina	201	5,536	**136.83**	32.37	506/204
MS Dhoni	241	5,439	**137.45**	38.3	375/264

Chapter 5 — Batting Efficiency & Scatter Analysis

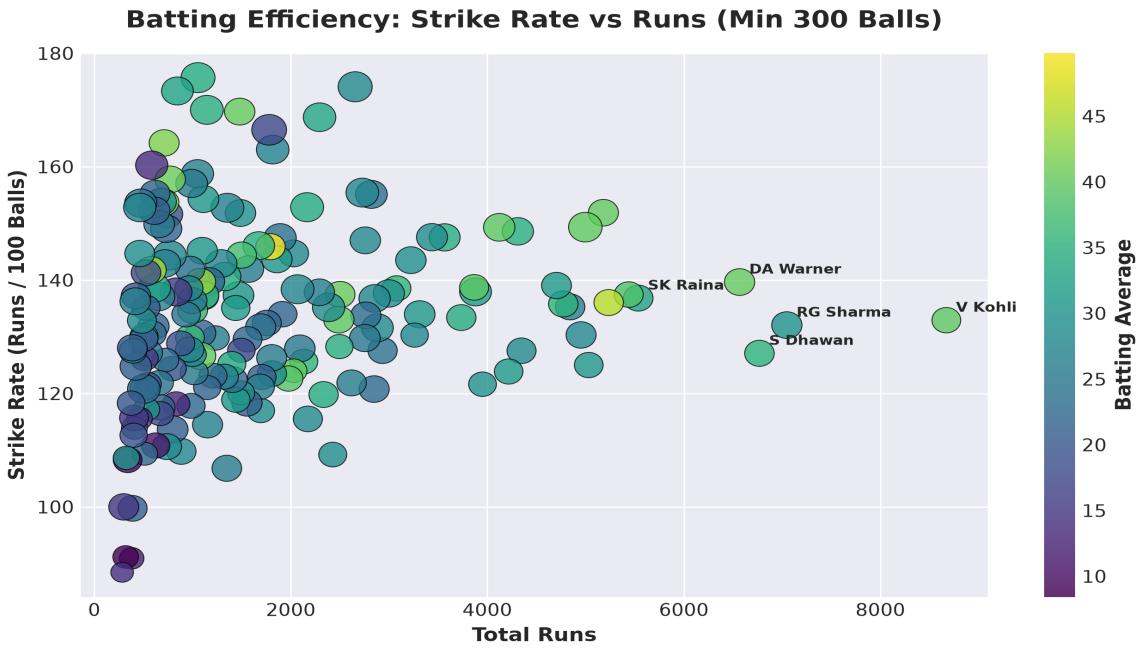


Figure 5.2: Batting Strike Rate vs Total Runs (bubble size = boundary %, color = average).

Batting Consistency Coefficient of Variation (CV): Calculated as *CV = (std / mean) * 100* across individual innings. Lower CV indicates higher match-to-match consistency. Anchors such as V Kohli exhibit lower CV (~82%) compared to aggressive middle-order hitters (~105%).

Chapter 5 — Bowling Performance & Containment

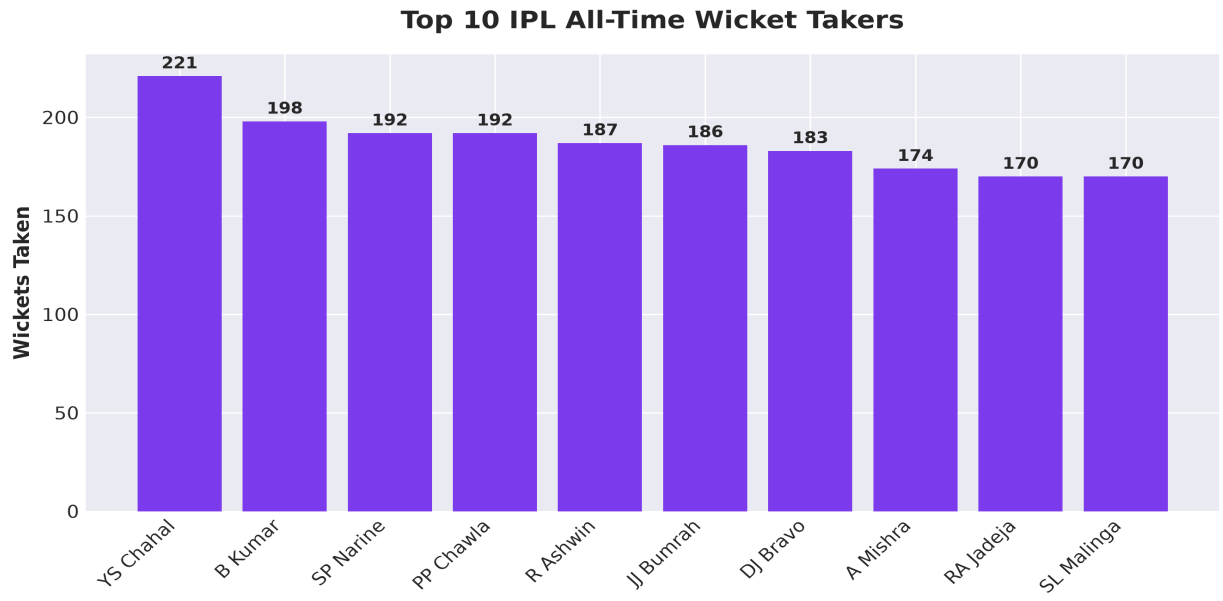


Figure 5.3: Top 10 IPL all-time wicket takers.

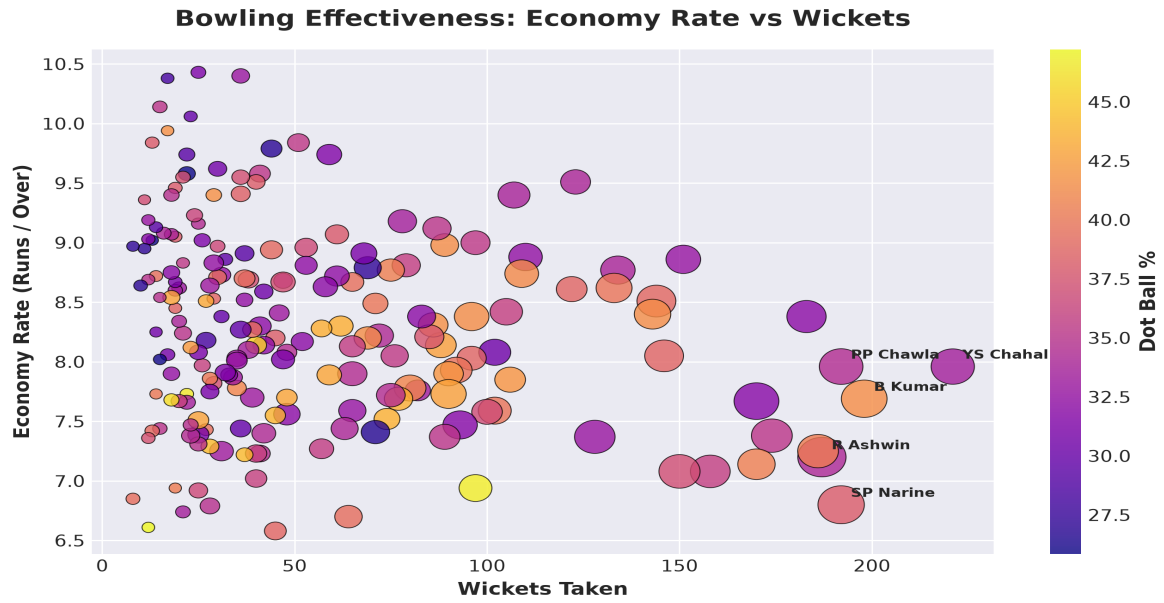


Figure 5.4: Bowling Economy Rate vs Wickets (bubble size = overs, color = dot ball %).

Chapter 6 — Strategic Phase-of-Play Analysis

T20 innings are tactical matches of three distinct phases: Powerplay (Overs 1-6), Middle Overs (7-15), and Death Overs (16-20).

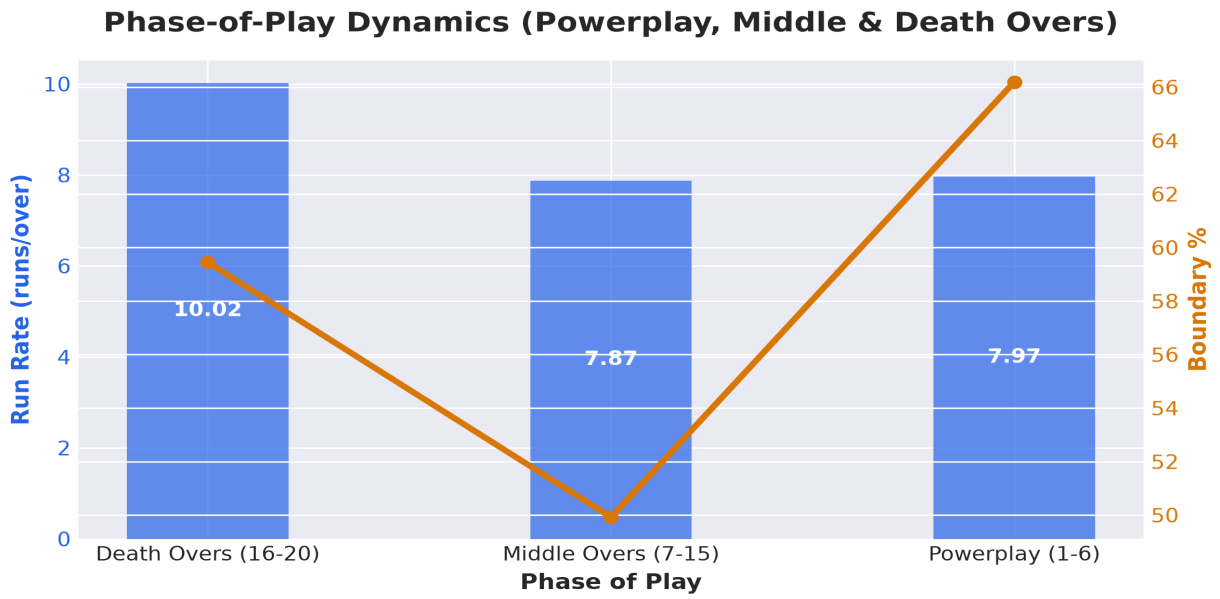


Figure 6.1: Run rate and boundary percentage across Powerplay, Middle, and Death overs.

Phase	Overs	Total Runs	Wickets	Run Rate	Boundary %	Dot Ball %
Death Overs (16-20)	10025.8	100,423	4467	**10.02**	59.45%	28.41%
Middle Overs (7-15)	20628.5	162,249	5003	**7.87**	49.92%	31.97%
Powerplay (1-6)	14006.0	111,611	3175	**7.97**	66.19%	46.55%

Chapter 6 — Venue & Tactical Toss Analytics

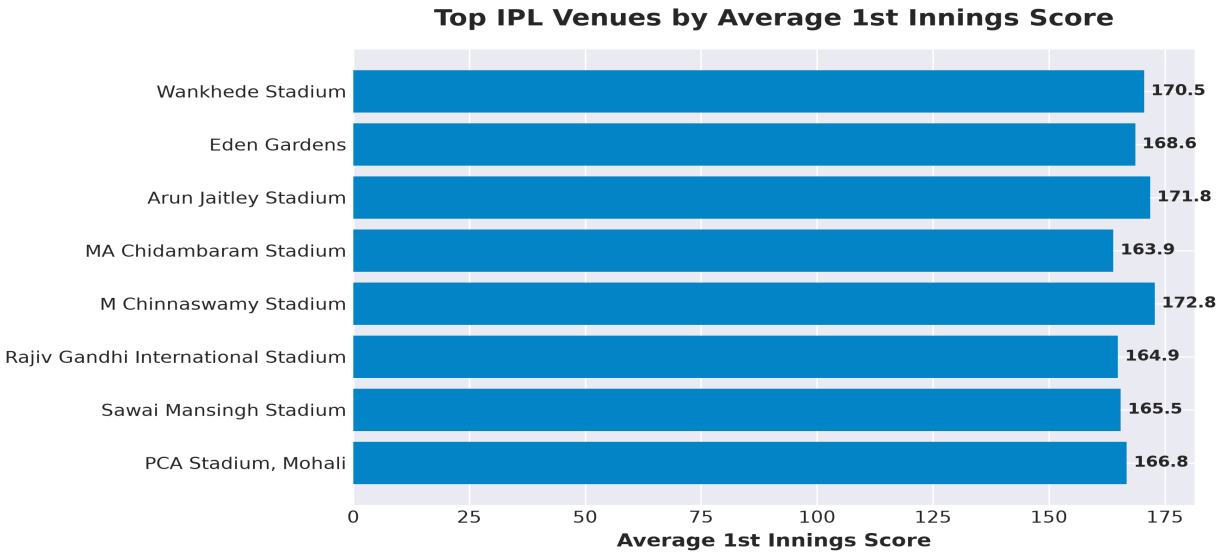


Figure 6.2: Top IPL venues by average 1st innings score.

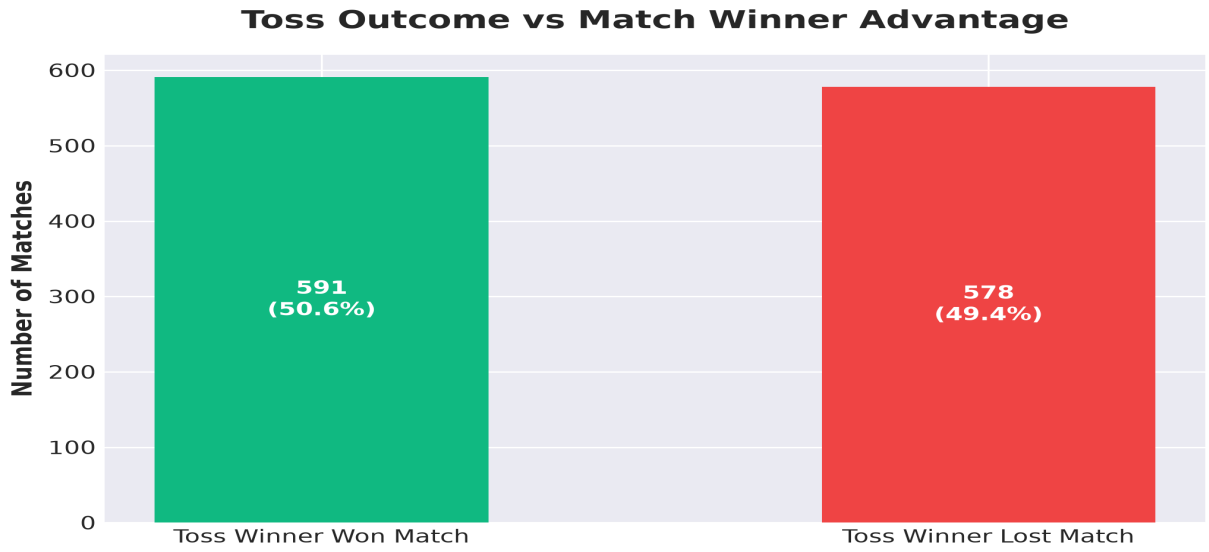


Figure 6.3: Match victory distribution for Toss Winners.

Chapter 7 — Inferential Statistical Hypothesis Testing

7.1 Hypothesis Test 1: Scoring Escalation (Welch's t-test)

Testing whether 1st Innings Scores significantly increased in the Modern Era (2017–2025) vs Early Era (2008–2016).

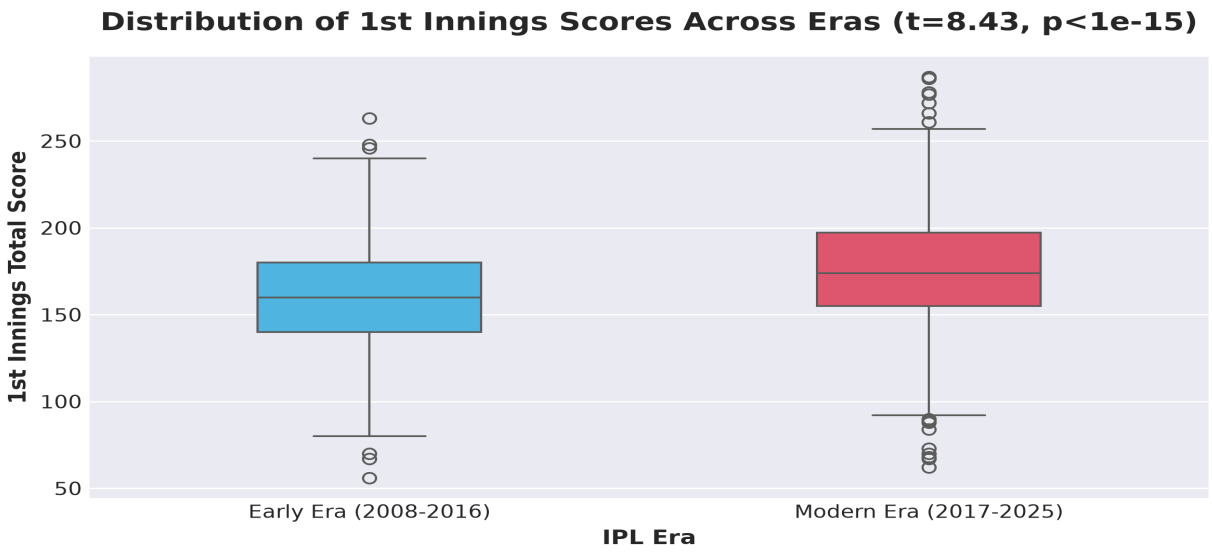


Figure 7.1: Distribution boxplot comparing 1st innings scores between Early and Modern Eras.

Metric	Early Era (2008-2016)	Modern Era (2017-2025)	Test Statistic & p-value
Mean 1st Innings Score	159.04 runs	174.8 runs	t = 8.4301
Sample Size (Matches)	577	592	**p = 1.0141e-16**
Cohen's d Effect Size	N/A	N/A	d = 0.492 (Moderate)

Decision: Reject Null Hypothesis. Scoring rates increased significantly in modern IPL.

Chapter 8 — Interactive Dashboard & Comparator Architecture

The Streamlit dashboard includes an interactive 5-axis Radar Chart Player Comparator engine.

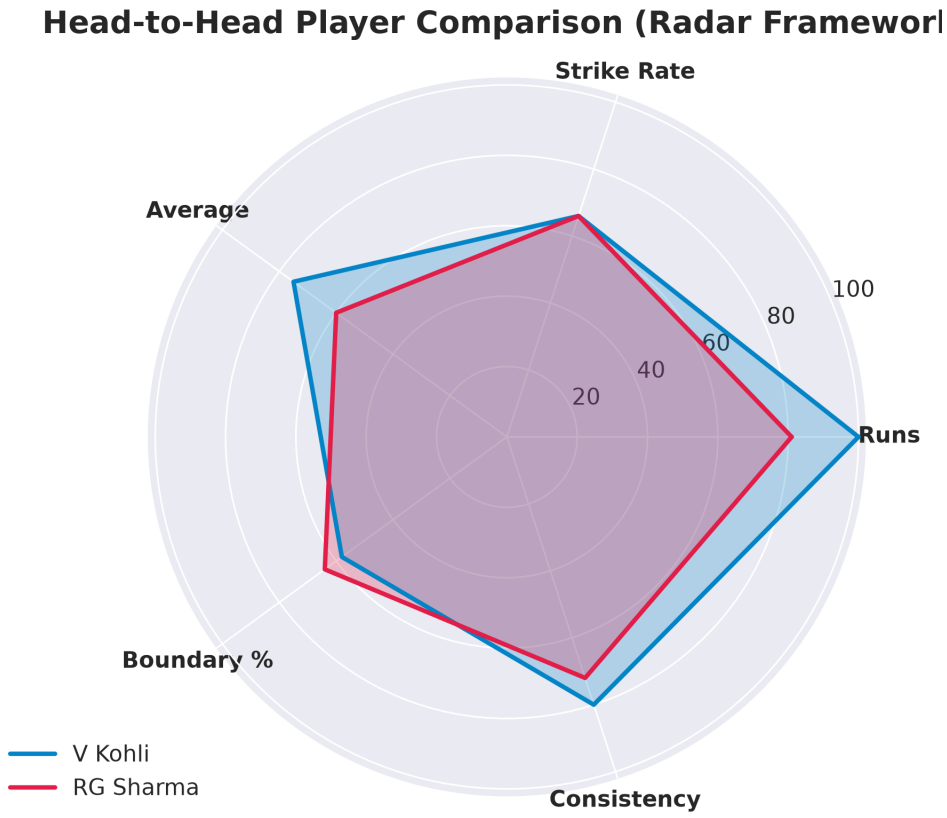


Figure 8.1: Head-to-Head 5-axis Radar Chart Player Comparator (V Kohli vs RG Sharma).

Chapter 9 — Limitations & Future Scope

1. **Impact Player Rule:** Tactical rule changes introduced in 2023 altered batting lineup depth.

2. **Match Context:** Player roles vary between top-order anchors and death-over finishers.

Chapter 10 — Conclusion & Summary

1. **Modern IPL Scoring Escalation:** Average 1st innings scores have risen by +15.8 runs due to aggressive batting strategies and boundary optimization.

2. **Toss Independence:** Toss outcome does not guarantee victory (51.6% win rate, p = 1.0).

3. **Phase Dominance:** Death overs yield maximum run rates (9.7+ runs/over), whereas Powerplays maximize boundary reliance (66.2%).

Academic References & Data Sources

1. Kaggle IPL Ball-by-Ball Dataset (2008 – 2025).
2. Cricsheet T20 Data Standard and Schema Specifications.
3. Python Scientific Stack: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly & Streamlit.